

CoPE Staff Day

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- PhD in mathematics at Aalto Uni. 2017
- Postdoc at UNSW Sydney 2018–2020
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Quasi-Monte Carlo methods

High-dimensional numerical integration

$$\int_{[0,1]^s} f(\mathbf{y}) \, d\mathbf{y} \approx \sum_{i=1}^n w_i f(\mathbf{t}_i)$$

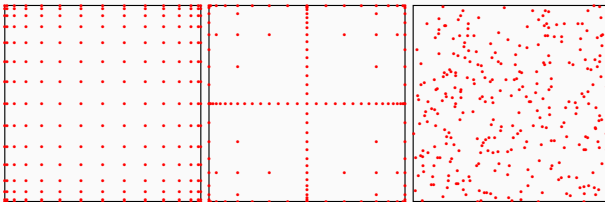


Figure 1: Tensor product grid, sparse grid, Monte Carlo nodes (not QMC rules)

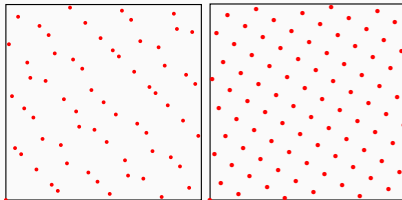


Figure 2: Sobol' points, lattice rule (examples of QMC rules)

Quasi-Monte Carlo (QMC) methods are a class of *equal weight* cubature rules

$$\int_{[0,1]^s} f(\mathbf{y}) \, d\mathbf{y} \approx \frac{1}{n} \sum_{i=1}^n f(\mathbf{t}_i),$$

where $(\mathbf{t}_i)_{i=1}^n$ is an ensemble of *deterministic* nodes in $[0, 1]^s$.

The nodes $(\mathbf{t}_i)_{i=1}^n$ are NOT random!! Instead, they are *deterministically chosen*.

QMC methods exploit the smoothness and anisotropy of an integrand in order to achieve better-than-Monte Carlo rates.

Lattice rules

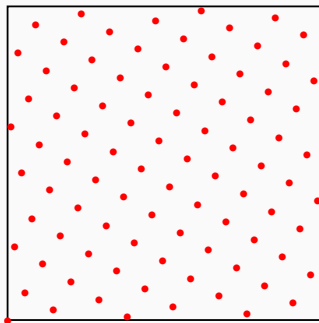
Rank-1 lattice rules

$$Q_{s,n}(f) = \frac{1}{n} \sum_{i=1}^n f(\mathbf{t}_i)$$

have the points

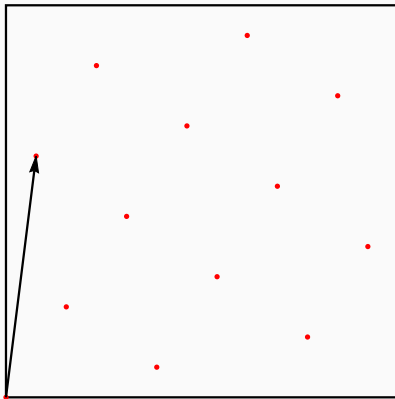
$$\mathbf{t}_i = \text{mod} \left(\frac{i\mathbf{z}}{n}, 1 \right), \quad i \in \{1, \dots, n\},$$

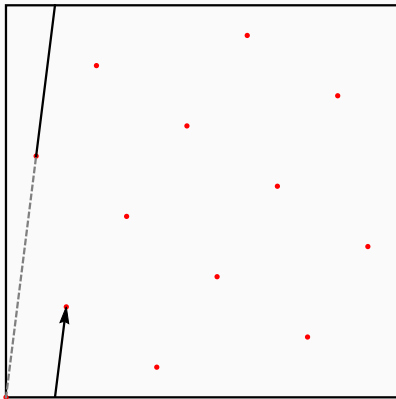
where the entire point set is determined by the *generating vector* $\mathbf{z} \in \mathbb{N}^s$, with all components *coprime* to n .

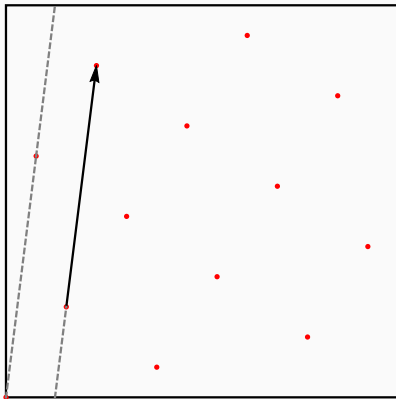


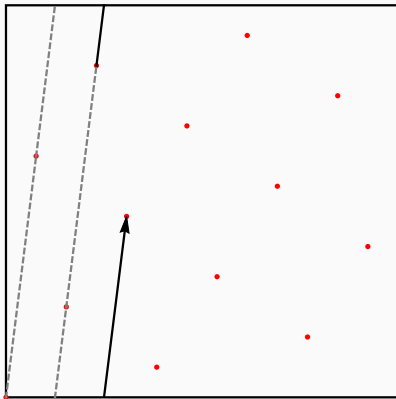
Lattice rule with $\mathbf{z} = (1, 55)$ and $n = 89$
nodes in $[0, 1]^2$

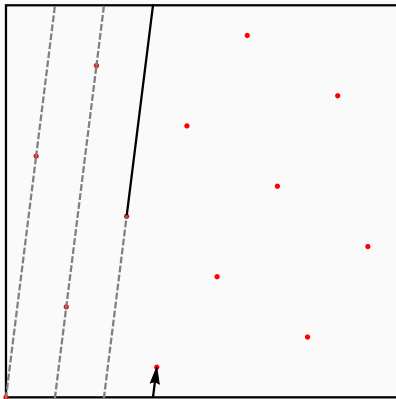
The quality of the lattice rule is determined by the choice of $\mathbf{z}$.

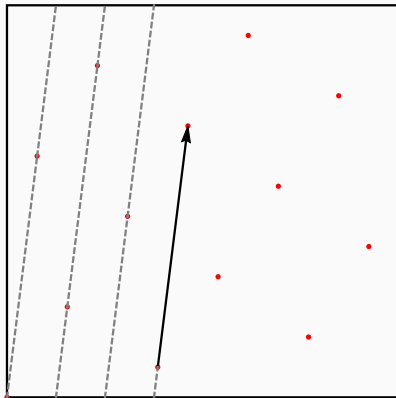


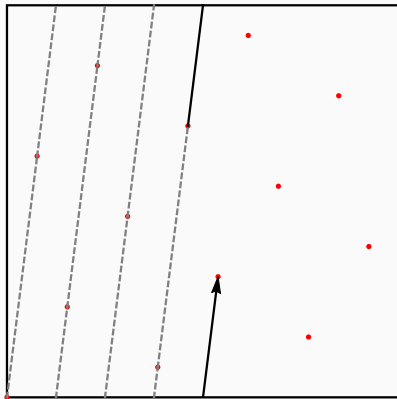


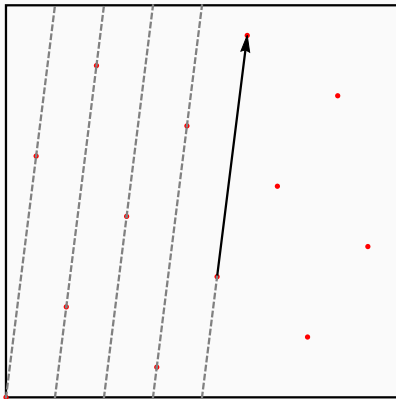


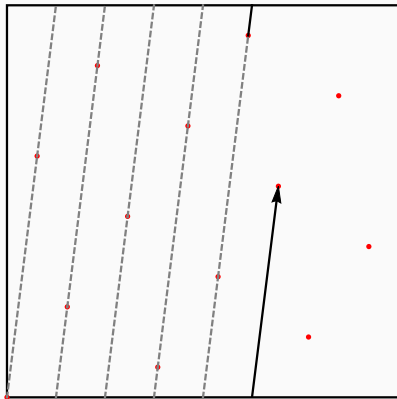


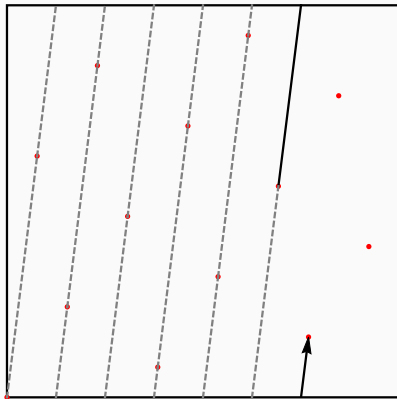


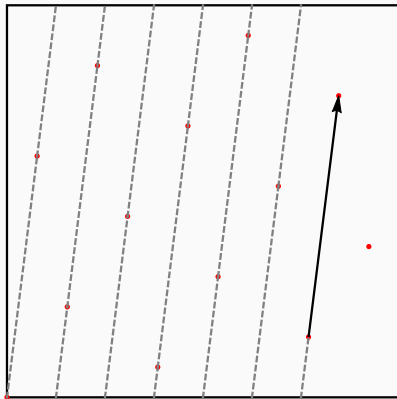


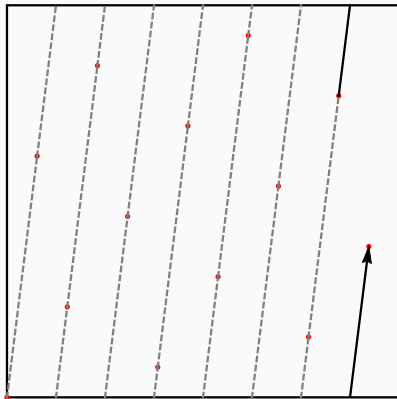












Some applications of QMC in PDE uncertainty quantification

Consider the elliptic PDE problem:

$$\begin{cases} -\nabla \cdot (a(\mathbf{x})\nabla u(\mathbf{x})) = f(\mathbf{x}) & \text{for } \mathbf{x} \in D, \\ +\text{boundary conditions.} \end{cases}$$

In practice, one or several of the material/system parameters may be uncertain or incompletely known and modeled as random fields:

- PDE coefficient a may be uncertain;
- Source term f may be uncertain;
- Boundary conditions may be uncertain;
- The domain D itself may be uncertain.

In forward uncertainty quantification, one is interested in assessing how uncertainties in the inputs of a mathematical model affect the output.

⇒ If the uncertain inputs are modeled as random fields, then the output of the PDE is also a random field. One may be interested in assessing the statistical response of the system, for example, the expectation or variance of the PDE solution (or some other quantity of interest thereof).

Uncertainty in groundwater flow

Risk analysis of radwaste disposal or CO₂ sequestration.

Darcy's law

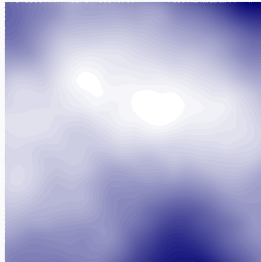
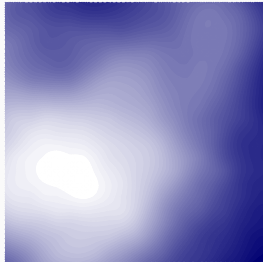
$$\mathbf{q}(\mathbf{x}) + \mathbf{a}(\mathbf{x})\nabla p(\mathbf{x}) = \mathbf{f}(\mathbf{x})$$

mass conservation law

$$\nabla \cdot \mathbf{q}(\mathbf{x}) = 0$$

in $D \subset \mathbb{R}^d$, $d \in \{1, 2, 3\}$

together with boundary conditions

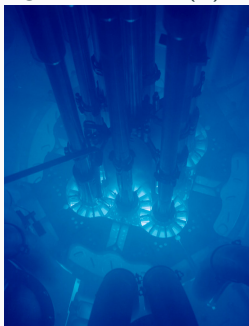


Uncertainty in $\mathbf{a}(\mathbf{x}, \omega)$ leads to uncertainty in $\mathbf{q}(\mathbf{x}, \omega)$ and $p(\mathbf{x}, \omega)$

Criticality problem for nuclear reactors

$$-\nabla \cdot \underbrace{\left(a(\mathbf{x}) \nabla u(\mathbf{x}) \right)}_{\text{diffusion}} + \underbrace{b(\mathbf{x})}_{\text{absorption}} u(\mathbf{x}) = \lambda \underbrace{c(\mathbf{x})}_{\text{fission}} u(\mathbf{x})$$

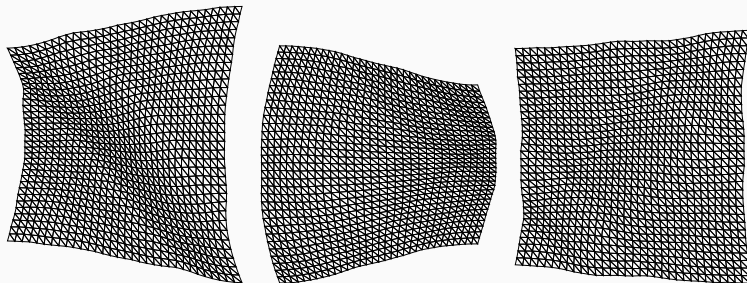
- The smallest eigenvalue $\lambda_1 \in \mathbb{R}$ measures *criticality* of a reactor.
- Eigenfunction $u_1(\mathbf{x})$ is the *neutron flux* at the point $\mathbf{x}$.



- $\lambda_1 \approx 1 \Rightarrow$ operating efficiently
- $\lambda_1 > 1 \Rightarrow$ not self-sustaining
- $\lambda_1 < 1 \Rightarrow$ supercritical

Source: Argonne National
Laboratory on Flickr

Domain uncertainty quantification

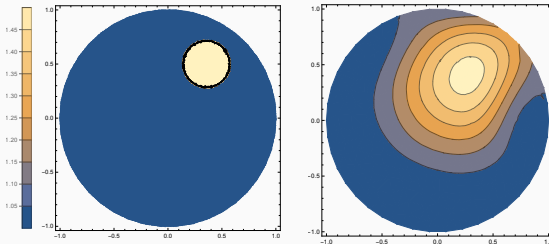
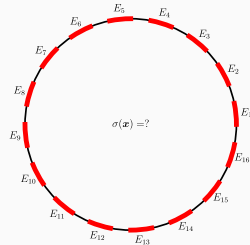


Three realizations of a random spatial domain

Electrical impedance tomography

Use measurements of current and voltage collected at electrodes covering part of the boundary to infer the interior conductivity of an object/body.

$$\begin{cases} \nabla \cdot (\sigma \nabla u) = 0 & \text{in } D, \\ \sigma \frac{\partial u}{\partial \mathbf{n}} = 0 & \text{on } \partial D \setminus \bigcup_{k=1}^L \overline{E_k}, \\ u + z_k \sigma \frac{\partial u}{\partial \mathbf{n}} = U_k & \text{on } E_k, \quad k \in \{1, \dots, L\}, \\ \int_{E_k} \sigma \frac{\partial u}{\partial \mathbf{n}} \, dS = I_k, & k \in \{1, \dots, L\}, \end{cases}$$



- We are always looking for new problems where we can apply QMC!
 - Do you have a problem where QMC might be useful? :)
- Recent trends: function approximation using QMC point sets, training neural networks over QMC point sets, superconvergence properties of smooth functions, extending QMC theory for nonsmooth functions, high-dimensional double integrals (design of experiments), ...

Thank you for your attention!